

Identification of Novel Syncytiotrophoblast Membrane Extracellular Vesicles Derived Protein Biomarkers in Preeclampsia: A Cross-Sectional Study

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Abstract

Background

Preeclampsia (PE), a multi-systemic hypertensive pregnancy disease that affects 2-8% of pregnancies worldwide, is a leading cause of adverse maternal and fetal outcomes. Current clinical PE tests have a low positive predictive value for PE prediction and diagnosis. The placenta notably releases extracellular vesicles from the syncytiotrophoblast (STB-EV) into the maternal circulation.

Objective

To identify a difference in placenta and STB-EV proteome between PE and normal pregnancy (NP), which could lead to identifying potential biomarkers and mechanistic insights.

Methods

Using ex-vivo dual lobe perfusion, we performed mass spectrometry on placental tissue, medium/large and small STB-EVs isolated from PE (n = 6) and NP (n = 6) placentae. Bioinformatically, mass spectrometry was used to identify differentially carried proteins. Western blot was used to validate the identified biomarkers. We finished our investigation with an in-silico prediction of STB-EV mechanistic pathways.

Results

We identified a difference in the STB-EVs proteome between PE and NP. Filamin B, collagen 17A1, pappalysin-A2, and scavenger Receptor Class B Type 1) were discovered and verified to have different abundances in PE compared to NP. In silico mechanistic prediction revealed novel mechanistic processes (such as abnormal protein metabolism) that may contribute to the clinical and pathological manifestations of PE.

Conclusions

We identified potentially mechanistic pathways and identified differentially carried proteins that may be important in the pathophysiology of PE and are worth investigating because they could be used in future studies of disease mechanisms and as biomarkers.

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eLife assessment

This study presents **valuable** findings that could be utilized for the identification of women at risk for preeclampsia prior to the onset of the disease. The novel aspect of this study lies in the utilization of exosomes with two different sizes. However, the data is **incomplete**: the patient population has not been well-defined, and the study only measured the proteins at a single time point.

Introduction

Preeclampsia (PE) is a significant cause of maternal and neonatal morbidity and mortality, affecting 2-8% of all pregnancies (Lisonkova & Joseph, 2013 [↗](#)). It is characterized by hypertension (systolic blood pressure ≥ 140 mmHg / diastolic pressure ≥ 90 mmHg), and either proteinuria (protein/creatinine ratio of ≥ 30 mg/mmol or more), or evidence of maternal acute kidney injury, liver dysfunction, neurological abnormalities, hemolysis, or thrombocytopenia, and/or fetal growth restriction. (“ACOG Practice Bulletin No. 202: Gestational Hypertension and Preeclampsia,” 2019; Brown et al., 2018 [↗](#)) Predicting or early detection of PE is thus of extreme importance to reduce the chance of long term complications but this has been challenging due to the limitations of current predictive models and biochemical tests (which lack in positive predictive value) (Zeisler et al., 2016 [↗](#)). The existing tests perform far better in ruling out rather than ruling in PE. They are also most effective shortly before the onset of the disease and within a specific time frame (1 or 2 weeks) rather than earlier in pregnancy (Thadhani et al., 2022 [↗](#)).

The pathophysiology of PE implicates the placenta. It is known that PE can occur in trophoblastic tumors (without the presence of a fetus); that PE is more common in multiple pregnancy (with greater placental mass) and that it has occurred in ectopic pregnancies (excluding the

involvement of the uterus)(Billieux et al., 2004 [↗](#); Hailu et al., 2017 [↗](#); C. W. G. Redman et al., 2022 [↗](#); Soto-Wright et al., 1995 [↗](#)). Finally, delivery of the placenta (irrespective of gestational age) is currently the only cure for the condition(C. Redman, 2014 [↗](#)).

The placental syncytiotrophoblast (STB) layer, the interface between the fetus and the mother which lies in direct contact with the maternal circulation. EVs, including STB-EVs, are membrane-bound and cell-derived particles that carry different cargos, including proteins, ribonucleic acid (RNA), deoxyribonucleic acid (DNA) and lipids(Raposo & Stoorvogel, 2013 [↗](#)). EVs are based on size into medium/large STB-EVs (MVs, 201–1000nm) or small STB-EVs (≤ 200 nm)(Dragovic et al., 2015b [↗](#)). The release of syncytiotrophoblast extracellular vesicles (STB-EVs) into the maternal circulation increases with gestation and is further elevated in PE. (Germain et al., 2007 [↗](#); Goswamia et al., 2006 [↗](#))

The presence of proteins, RNA species and DNA in EVs combined with their constitutional release as inter-cellular signaling moieties means they also have potential in disease prediction and diagnosis. In this study, we performed proteomic analysis of placenta and STB-EVs in PE and NP to identify potential placenta-derived biomarkers. We also conducted in silico analysis on the proteomes to detect potential molecular targets, mechanisms, and processes of PE.

Methods

Ethics approval and patient information

Oxfordshire Research Ethics Committee C (07/H0606/148) approved this study. Normal pregnancy was a healthy singleton pregnancy with normal maternal blood pressure. PE was defined as new (after 20 weeks) systolic blood pressure 140 mmHg or diastolic pressure 90 mmHg, proteinuria (protein/creatinine ratio of 30 mg/mmol or more), maternal acute kidney injury, liver dysfunction, neurological features, haemolysis, thrombocytopenia, and/or foetal growth restriction. This study included only early-onset PE patients (diagnosed before 34 weeks gestation). After informed consent, placenta was obtained from women undergoing elective caesarian section without labor were collected.

Placenta sample preparation and syncytiotrophoblast membrane extracellular vesicles (STB-EVs) enrichment

Placenta biopsies were obtained adjacent to the perfused lobe and immediately frozen at -80°C pending transfer to the Target Discovery Institute (Oxford) for proteomic analysis. We obtained STB-EVs via placental perfusion as previously described(Dragovic et al., 2015a [↗](#)). Our STB-EV enrichment and categorization process has been deposited on EV Track (<http://www.EVTRACK.org> [↗](#)], EV-TRACK ID: EV 220157) with a score of 78% (the average score on EV track for 2021 is 52 %). Full details can be found in the supplemental data.

Characterization of syncytiotrophoblast membrane extracellular vesicles (STB-EVs)

Enriched STB-EVs were resuspended in filtered phosphate buffered saline (fPBS) and characterized with bicinchoninic acid (BCA) assay (for protein concentration) and nanoparticle tracking analysis (NTA) (for particle number and size profile). We also phenotyped the STB-EVs with transmission electron microscopy (for morphology), flow cytometry (BD Biosciences, LSRII), and western blot (for immunophenotyping). Flow cytometric analysis was performed using antibodies to placental alkaline phosphatase – PLAP-(to confirm syncytiotrophoblast origin), CD 41 (to identify co-isolated platelet EVs, CD235 a/b (to identify co-isolated red blood cell EVs), HLA class I and II (to identify co-isolated white blood cell EVs). Western blot was probed for placental

alkaline phosphatase (PLAP [1.667 mg/ml] at 1:1000 dilution in house antibody), the known EV markers CD 63 ([200ug/ml] at 1:1000 dilution, Sc-59286, Santa Cruz Biotechnology), ALIX ([200ug/ml] at 1:1000 dilution, Sc-53538, Santa Cruz Biotechnology) and the known negative EV marker Cytochrome C ([200ug/ml] at 1:1000 dilution, Sc-13156, Santa Cruz Biotechnology) as recommended by the international society for extracellular vesicles (ISEV) and subsequently incubated with the corresponding secondary antibody anti-mouse, or anti-rabbit polyclonal goat immunoglobulins/HRP (at 1: 2000 dilution, Dako UK Ltd, Cambridgeshire UK). Details of nanoparticle tracking analysis and transmission electron microscopy can be found in the supplemental data.

Sample preparation for Mass Spectrometric analysis and bioinformatic analysis of proteomic data from placenta tissue, medium/large, and small STB-EVs

STB-EVs and placental tissue samples were processed for liquid chromatography mass spectrometry (LC-MS) (Target Discovery Institute, Oxford). Raw label-free quantitation (LFQ) data was imported and analyzed in Perseus (Max Planck Institute of Biochemistry). Functional enrichment analysis was done with g: Profiler(Raudvere et al., 2019 <https://biit.cs.ut.ee/gprofiler/gost>) The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD031953.

Bioinformatic analysis of proteomic data from placenta tissue, medium/large and small STB-EVs

Persus (Max Planck Institute of Biochemistry) was used for the analysis, along with the accompanying documentation and tutorials. To remove invalid data, we pre-processed the raw data by log transforming and filtering. Missing values were imputed at random from a normal distribution using the following parameters: width = 0.3 and downshift = 1.8. To ensure conformity to the normal distribution, the underlying distribution was visually inspected with a histogram before and after missing data imputation. The data was then transformed further by deducting each (transformed) value from the highest occurring protein expression value.

Principal component analysis (PCA), heatmaps, and Pearson correlation matrices were used to further investigate the data. The data was analyzed using the correlation index and hierarchical clustering. A two-sample independent student t-test was used to assess differential expression. Multiple testing was corrected using permutation-based false discovery rate (FDR) with the following parameters, with significance set at less than 0.05.

Western Blotting

To further characterize and immune-phenotype, we performed western blots on placental homogenates and STB-EV pellets. The relevant primary antibodies were used to probe all STB-EVs (as discussed in the main article). An equal amount of protein (20 micrograms) was mixed with 4 X Laemmli buffer ((180 mM Tris-Cl (pH 6.8), 6% SDS, 30% glycerol, 0.3% 2-mercaptoethanol, and 0.0015% bromophenol blue, BioRad)). After heating the sample mix for ten minutes at 70°C, an equal volume of the sample mix was loaded. Electrophoresis was performed at 150 V for 1.5 hours in a Novex minicell tank (Invitrogen, UK) filled with NuPAGE™ MOPS SDS running buffer under reducing (for PLAP, Cytochrome C, ALIX, SR-BI, Filamin B, PAPP-A2, Collagen 17 A1) and non-reducing (for CD 63) conditions on NuPAGE™ 4-12% Bis-Tris Gel 1.0 mm x 10 well gels (Invitrogen by Thermo Fisher Scientific) (Novex by Life Technologies). As a protein size marker, Precision plus protein™ dual color standards (Bio-Rad Laboratories Ltd, Hertfordshire, UK) were used. Following protein separation, the proteins were transferred to a polyvinylidene difluoride (PVDF) membrane (Bio-Rad).

In a Novex semi-dry transfer apparatus, the PVDF membrane and gel were sandwiched between four filter paper sheets pre-soaked in anode one buffer solution (300 mM Tris, 10% methanol pH 10.4) and two filter paper sheets pre-saturated with anode two buffer solution (25 mM Tris, 20% methanol pH 10.4) at the bottom and three filter paper sheets pre-soaked in cathode buffer solution (25 mM Tris, 40 mM (Life Technologies, UK). For 45 minutes, the transfer was run at 25 V. The membranes were blocked for one hour with 5% Blotto (2BScientific) in 0.1% TBST (Tris-buffered saline (20 mM Tris, 137 mM NaCl, pH 7.6) and 1% Tween-20, Sigma), then incubated overnight with the appropriate antibody.

Antibodies used were SR-BI (ab52629 [0.35 µg/µl] 1:1000 dilution, monoclonal Rabbit, Abcam), Filamin B (GTX387 [1.64 µg/µl] 1: 1000 dilution, monoclonal mouse, Insight Biotechnology), PAPP-A2 (ab59100 [1 µg/µl] 1: 2000 dilution polyclonal Rabbit, Abcam), Collagen 17 A1(ab28440 [1 µg/µl] 1: 2000 dilution polyclonal Rabbit, Abcam), and for characterization, placental alkaline phosphatase (PLAP [1.667 µg/µl] at 1:1000 dilution in house antibody), CD 63 ([200 µg/µl] at 1:1000 dilution, Sc-59286, Santa Cruz Biotechnology), Alix ([200µg/µl] at 1:1000 dilution, Sc-53538, Santa Cruz Biotechnology), and Cytochrome C ([200 µg/µl] at 1:1000 dilution, Sc-13156, Santa Cruz Biotechnology) as recommended by the international society for extracellular Vesicles (ISEV).

After an overnight incubation and three five-minute TBST washes, the membranes were incubated for an hour at room temperature with the corresponding secondary antibody, anti-mouse (P044701-2) or anti-rabbit (P044801-2) polyclonal goat immunoglobulins/ horseradish peroxidase (HRP) (1: 2000 dilution, Dako UK Ltd, Cambridgeshire UK) and rinsed three times in TBST. The membranes were developed using a gel documentation system (G-Box, Syngene, Cambridge UK) running GeneSys (version 1.5.0.0, Syngene) and a chemiluminescence film (Amersham Hyperfilm™ ECL (enhanced chemiluminescence; GE Healthcare Limited, Buckinghamshire UK) to obtain the band intensity for each lane. The fold change (FC) between normal and preeclampsia samples was calculated by normalizing the comparative expression analysis to the total protein loaded (using Amido Black stain) 17,18. The normalized densitometric values were statistically tested using a one-tailed Student t-test, with significance set at less than 0.05. **Tables 1** and **2** show the reagents and antibodies used in western blot analysis, respectively. Details of western blot for STB-EV characterization and proteomic validation experiments can be seen in the supplemental material.

Results

Patient demographics and clinical characteristics

PE mothers (**Table 1**), as expected, had a significantly higher average systolic (178.83 mmHg, $P < 0.001$) and diastolic (109.17 mmHg, $P < 0.001$) blood pressure compared to normal pregnant mothers average systolic (129.50 mmHg) and diastolic (67.00 mmHg). PE women are also significantly more likely to deliver prematurely (PE = 32.00 weeks gestation NP = 39.17 weeks gestation, $P < 0.001$) and have proteinuria (PE = 2.58 pluses on urine dipstick NP 0 pluses on urine dipstick, $P < 0.001$) compared to normal pregnancy. Finally, PE babies significantly weighed less than normal babies (PE = 1515.83g NP = 3912.50 g, $P < 0.001$). Surprisingly, we found no significant difference in body mass index, the gender of the child, and maternal age.

Characterization of syncytiotrophoblast membrane extracellular vesicles (STB-EVs)

Flow cytometry (**Figure 1**) was showed that many detected events ($83-85 \pm 8.0-8.3\%$) were negative for CD 235a (Red blood cells), CD41 (platelets) and HLA-I and II (white blood cells) (**Figure 1B**) while $92 \pm 0.9\%$ (**Figure 1D** and **1G**) of detected events were PLAP⁺ extracellular vesicles (BODIPY FL N-(2-aminoethyl)-maleimide (bioM) and placental alkaline phosphatase

Characteristics	Sub-Classification	Normal		P
		Pregnancy	Preeclampsia	Value
Sample size		6	6	
Maternal age years (mean (SD))		34.50 (5.39)	36.33(4.13)	0.524
Systolic blood pressure mmHg (mean (SD))		129.50 (4.93)	178.83(12.56)	<0.001
Diastolic blood pressure mmHg (mean (SD))		67.00 (6.20)	109.17(9.85)	<0.001
Body mass index kg/m-2 (mean (SD))		29.92(9.10)	31.25(11.12)	0.825
Proteinuria plus(es) (mean (SD))		0	2.58(1.20)	<0.001
Gestational age at delivery in weeks (mean (SD))		39.17 (0.98)	32.00(3.52)	0.001
Birth weight grams (mean (SD))		3912.50 (730.40)	1515.83(600.57)	<0.001
Intrauterine growth restriction (IUGR) =				
Yes (%)		0(0)	6(100)	0.004
Male new-born gender (%)		2(33.3)	2(33.3)	1.000

Table 1.

General descriptive statistics of sample population

Protein	Symbol	LFC	Adjusted P. Value
Placenta			
Isoform HMG-R of High mobility group protein HMG-I/HMG-Y	HMGA1	3.14	0.03
Fibrinogen-like protein 1	FGL1	2.49	0.04
Isoform 1 of Kinesin-like protein KIF2A	KIF2A	1.47	0.02
Chloride intracellular channel protein 3	CLIC3	1.09	0.05
Mesencephalic astrocyte-derived neurotrophic factor	MANF	1.02	0.03
Ig kappa chain V-I region AG	KV101	-1.02	0.05
Haloacid dehalogenase-like hydrolase domain-containing protein 2 (Fragment)	K7EJQ8	-1.23	0.05
Ig kappa chain C region	IGKC	-1.27	0.05
Alpha-1B-glycoprotein	A1BG	-1.34	0.05
Serum paraoxonase/arylesterase 1	PON1	-2.18	<0.01
Medium/Large STB-EVs			
Collagen alpha-1(XVII) chain	COL17A1	4.79	<0.01
Isoform 2 of Filamin B	FLNB	3.09	<0.01
Tumor necrosis factor alpha-inducible protein	TNFAIP2	3.03	<0.01
Minor histocompatibility antigen	HMHA1	1.59	0.03

Table 2

**Top ten differentially expressed Proteins between normal and preeclampsia
placentas (bold font), medium/large STB-EVs (normal font), small STB-EVs(italics)**

Bridging integrator 2	BIN2	2.33	0.02
Sodium/potassium-transporting ATPase subunit beta-1	ATP1B	1.99	0.02
ERO-1 like protein alpha	ERO1L	1.97	<0.01
Endoglin	ENG	1.82	<0.01
Methylthioribose-1-phosphate isomerase (Fragment)	MRI1	-1.96	0.04
Sodium-dependent phosphate transporter 1	S20A1/SLC17A1	-5.24	0.04
Small STB-EVs			
<i>Solute carrier family 2, facilitated glucose transporter member 11</i>	<i>SLC2A11</i>	<i>8.91</i>	<i>0.01</i>
<i>V-type proton ATPase 16 kDa proteolipid subunit</i>	<i>VATL</i>	<i>7.25</i>	<i>0.02</i>
<i>Pappalysin-2</i>	<i>PAPP2</i>	<i>3.76</i>	<i>0.02</i>
<i>ERO1-like protein alpha</i>	<i>ERO1A</i>	<i>3.29</i>	<i>0.02</i>
<i>Isoform 2 of Creatine kinase U-type, mitochondrial</i>	<i>KCRU</i>	<i>3.20</i>	<i>0.02</i>
<i>SCARB1 protein</i>	<i>B7ZKQ9</i>	<i>2.95</i>	<i>0.05</i>
<i>Isoform 2 of Filamin-B</i>	<i>FLNB</i>	<i>2.92</i>	<i>0.02</i>
<i>Solute carrier organic anion transporter family member 2A1</i>	<i>SO2A1</i>	<i>2.86</i>	<i>0.02</i>
<i>Isoform 2 of Oligaccharyltransferase complex subunit TC</i>	<i>OSTC</i>	<i>2.82</i>	<i>0.02</i>
<i>SCY1-like protein 2</i>	<i>SCYL2</i>	<i>2.71</i>	<i>0.02</i>

Table 2 (continued)

<i>Sodium-dependent phosphate transporter 1</i>	<i>S20A1</i>	<i>-2.93</i>	<i>0.01</i>
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Table 2 (continued)

(PLAP) double-positive). Detergent treatment with NP-40 confirmed that majority (99%) of our samples were vesicular our samples were largely vesicular since only $0.1 \pm 0.12\%$ of BODIPY FL N-(2-aminoethyl)-maleimide and PLAP double-positive events were detected (a reduction of 99%) (Figure 1E and 1H).

Transmission electron microscopy on 10K STB-EV pellets (Figure 2B and 2C) and 150K STB-EVs (Figure 2E and 2F) in our sample preparation showed the typical cup-shaped morphology of extracellular vesicles on transmission electron microscopy (TEM) within the appropriate size range. Western blot confirmed they express the classic syncytiotrophoblast membrane marker, placenta alkaline phosphatase (PLAP), the extracellular vesicle markers ALIX and CD 63. In addition, they lacked the negative EV marker cytochrome C (Figure 2G). Nanoparticle tracking analysis (NTA) confirmed the homogeneity of the 150K STB-EV pellets (small STB-EVs) (Figure 2A) with a modal size of (205.8 ± 67.7) nm and the heterogeneity of the 10K STB-EV pellets (medium/large STB-EVs) (Figure 2D) with a size range of (479.4 ± 145.6) nm.

Differentially carried proteins (DCPs) in Placenta homogenate, Medium/Large STB-EVs and Small STB-EVs in Preeclampsia versus normal pregnancies

In total, using mass spectrometry, there fifteen (15) proteins in the placenta, three hundred and four (304) in m/lSTB-EVs, and seventy-three (73) in sSTB-EVs were differentially expressed between preeclampsia (PE) and normal pregnancy (NP).

In the placenta (Table 2), isoform HMG-R of High mobility group protein (HMGA1), Fibrinogen-like protein 1 (FGL1), isoform 1 of Kinesin-like protein (KIF2A), Ig kappa chain C region (IGKC) were the most abundant proteins based on fold change. Concomitantly, serum paraoxonase/arylesterase 1 (PON1) and alpha-1B-glycoprotein (AIBG) were the least abundant proteins.

For m/lSTB-EVs (Table 2), the most differentially abundant proteins were the collagen alpha-1(XVII) chain (COL17A1), isoform 2 of Filamin-B (FLNB), tumor necrosis factor-alpha-induced protein 2 (Fragment) (TNFAIP2) based on fold change. In contrast, the least differentially abundant proteins were sodium-dependent phosphate transporter 1 (SLC20A1), methylthioribose-1-phosphate isomerase (Fragment) (MRI1), prostatic acid phosphatase (Fragment) (ACPP) based on fold change.

Finally, the sSTB-EVs (Table 2) had solute carrier family 2, facilitated glucose transporter member (SLC2A11), v-type proton ATPase 16 kDa proteolipid subunit (ATP6V0C), pappalysin 2 (PAPP-A2) as the most differentially abundant by fold change while sodium-dependent phosphate transporter 1 (SLC20A1) and isoform 2 of ADP-ribosylation factor GTPase-activating protein (ARFGAP3) were the least differentially abundant. There were 25 differentially carried proteins (DCPs) including filamin B (FLNB), ERO1-like protein alpha (ERO1A), endoglin (EGLN), pappalysin-2 (PAPP-A2), siglec6 (SIGL6) shared between m/lSTB-EVs and sSTB-EVs and only one protein, isoform 1 of kinesin-like protein (KIF2A) shared between the placenta and the m/l STB-EVs. Only one protein chloride intracellular channel protein 3 (CLIC3), was found in all three sample sub-types (Table 3).

Validation of select proteins in the placenta homogenate, medium/Large STB-EVs, and small STB-EVs

We combined all the DCPs identified from the placenta, m/lSTB-EVs, and sSTB-EVs and selected proteins to validate based on fold change and our placenta specificity or enrichment as previously described above in the methods section. These were pappalysin A2 (PAPP-A2), collagen 17A1 (COL17A1), filamin B, and scavenger receptor class B type I (SR-BI/ SCARB1). Validation was performed by western blot on all sample sub-types.

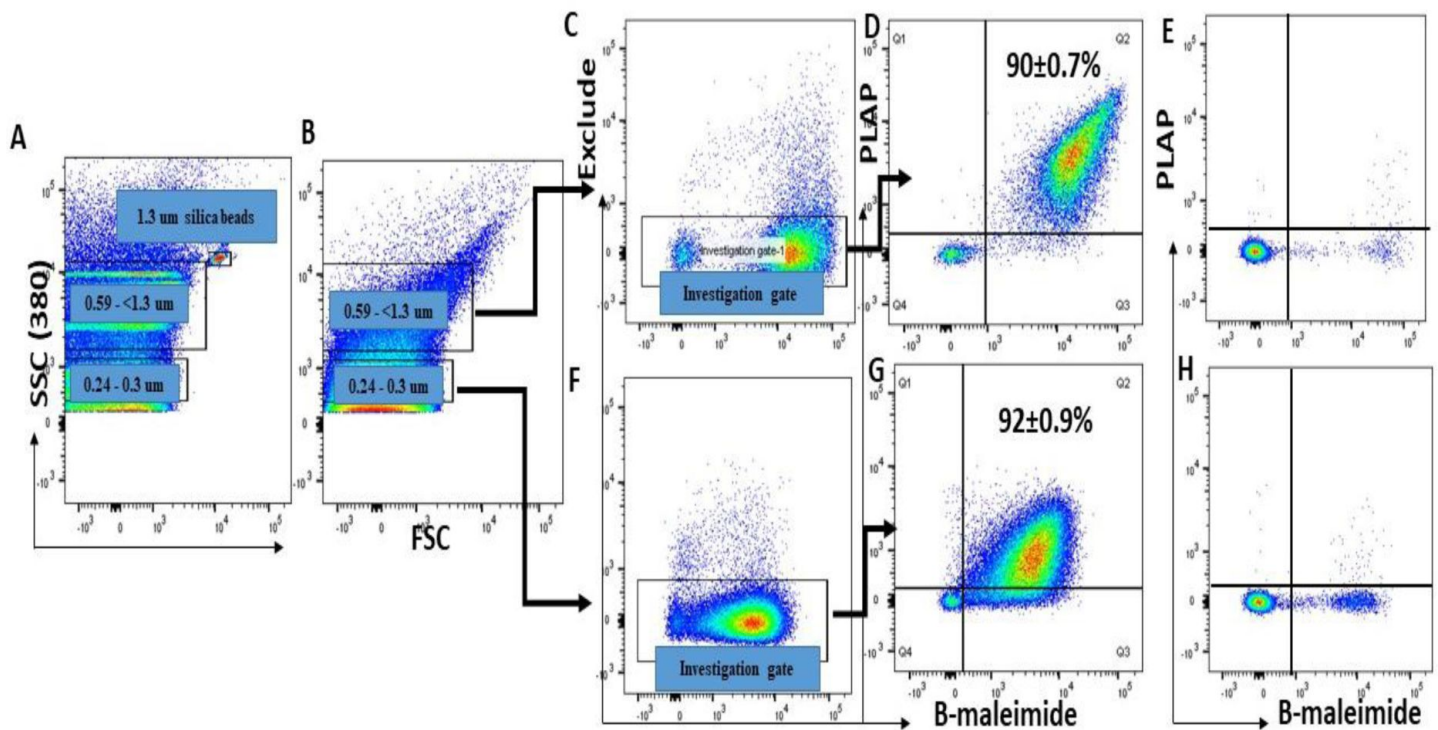


Figure 1.

Flow analysis of medium/large STB-EVs in the 10K STB-EV pellet. Apogee beads mix were used to set the flow machine's light scatter resolution to 0.59-1.3 mm and 0.24-1.3

(A) mm silica beads. Figures B show the application of SSC and FSC PMTVs as determined by apogee beads mix for the analysis of m/ISTB-EVs in the 10K pellet. An investigation gate was created to include only medium/large EVs negative for non-placental markers (C & F). Extracellular vesicles from the investigation gate were further analyzed for staining by bioM and expression of PLAP (D & G). Figure E and H shows that the bioM⁺ PLAP⁺ EVs were sensitive to detergent treatment. The percent of B-Maleimide⁺ PLAP⁺ from the 0.59-1.3 micron gate is consistent under both SSC conditions.

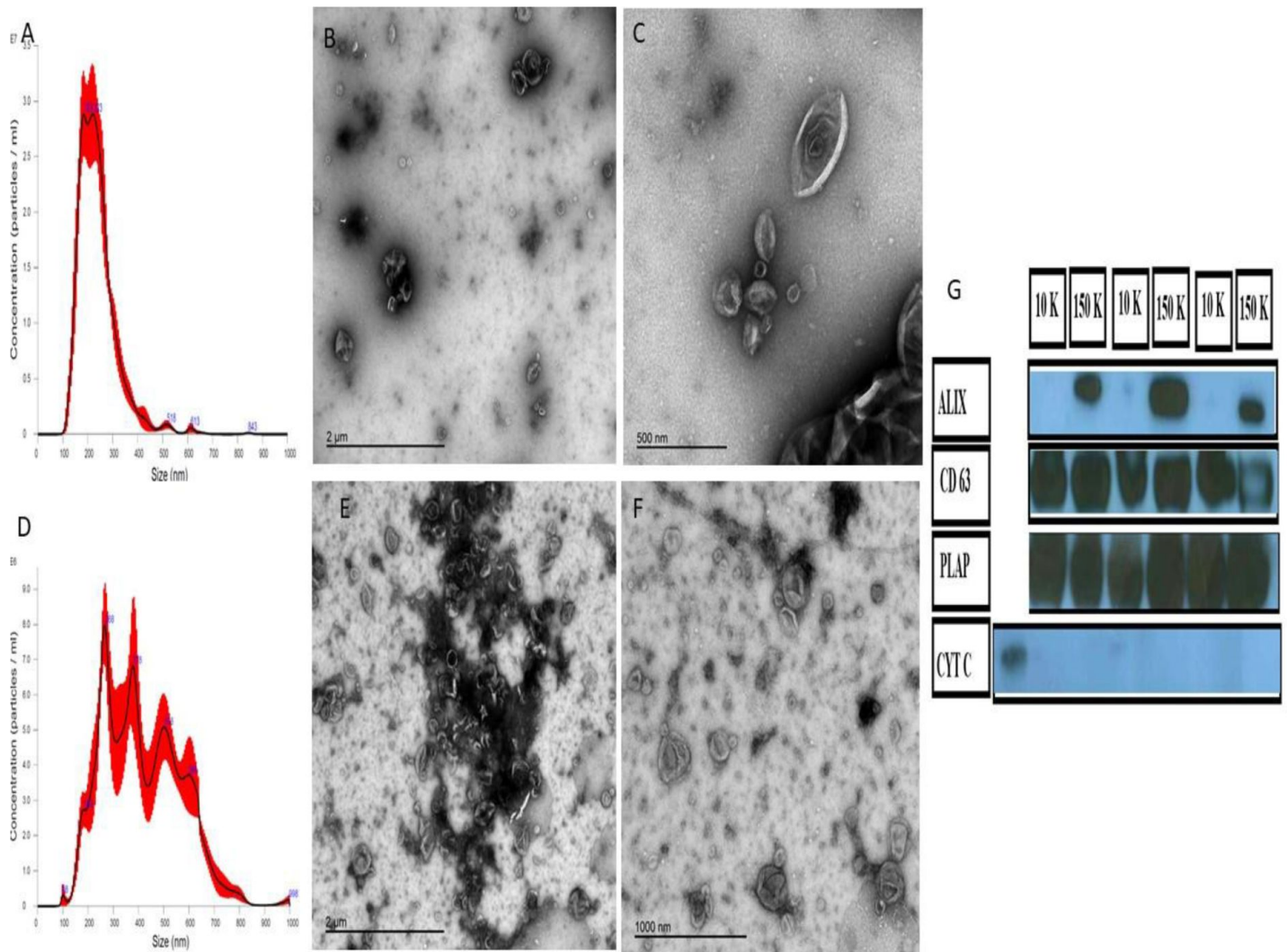


Figure 2.

Results of STB-EV characterization. **Figure 2A** and **2D** show the NTA results of S STB-EVs (2D) and m/ISTB-EVs (2A). **Figure 2B**, **2C**, **2E** and **2F** displays representative transmission electron microscopy (TEM) images with wide view (2B and 2E), medium/large STB-EVs (2C), and Small STB-EVs (2F).

Sample types	Differentially expressed proteins (DEPs)
m/ISTB-EVs and Placenta	Kinesin-like protein
m/ISTB-EVs, sSTB-EVs and Placenta	Chloride intracellular channel protein 3
m/ISTB-EVs and sSTB-EVs	Filamin B
	Endoplasmic Reticulum Oxidoreductase Alpha
	X-linked retinitis pigmentosa GTPase regulator interacting protein 1
	Protein NDRG1
	Collagen alpha 1(XVII) chain
	Monocyte differentiation antigen CD14
	Trophoblast glycoprotein
	Heat shock-related 70 kDa protein 2
	Endoglin
	Hippocalcin-like protein 1
	SCARB1 protein
	Importin subunit alpha 7
	Protein BCAP
	Endothelial protein C receptor
	Caveolin
	Pappalysin-2

Table 3.

List of overlapping differentially carried proteins in the placenta, medium/large STB-EVs and small STB-EVs

Phosphatidylinositol 3-kinase regulatory subunit alpha
Keratin, type I cytoskeletal 19
Siglec6
Protein disulfide-isomerase
Amine oxidase [flavin-containing] A
Copine-3
Dolichyl-diphosphooligosaccharide
Sodium-dependent phosphate transporter 1
Annexin A4

Table 3. (continued)

In the placenta (Figure 3A and 3D), *PAPP-A2* (FC = 6.39, P Value = 0.0001) and *SR-BI* (FC = 1.72, P Value = 0.04) were all differentially abundant in PE. *Filamin B* (FC = 2.59, P Value = 0.12) was differentially abundant in PE but not significant while *COL17A1* was undetectable. In medium/large STB-EVs (Figure 3B and 3E), pappalysin A2 (FC = 5.09, P Value = 0.004), *collagen 17A1* (FC = 71.21, P Value = 0.002), *filamin B* (FC = 7.60, P Value = 0.014) and *scavenger receptor class B type I* (FC = 2.28, P Value = 0.018) were all significantly differentially abundant in PE compared to normal. In sSTB-EVs (Figure 3C and 3F), only *filamin B* (FC = 7.73, P Value = 0.003) and *SR-BI/SCARB1* (FC = 1.60, P Value = 0.002) were significantly differentially abundant while *COL17A1* (FC = 12.79, P Value = 0.07) and *PAPP-A2* (FC = 1.48, P Value = 0.11) were differentially abundant but not significantly so.

Functional enrichment of differentially carried proteins (DCPs) in preeclampsia (PE)

We performed a functional enrichment analysis on the list of differentially carried proteins in placenta tissue, medium/large STB-EVs, and small STB-EVs to help better understand their role in preeclampsia (PE). The three sample sub-types did not have overlapping gene ontology terms or KEGG pathways. The top biological processes overrepresented in the placenta (Table 3 [↗](#)) were processes that involve *neurotransmitter secretion and transport*. In comparison, the enriched biological processes in the m/lSTB-EVs (supplemental table 3 [↗](#)) involved *responses to hypoxia*. Finally, *posttranslational protein modification processes* were enriched among small STB-EVs (supplemental table 3 [↗](#)). *Neurotrophin signaling pathway*, *spinocerebellar ataxia*, and *protein processing pathway* were among the over enriched KEGG pathways in the placenta, m/lSTB-EVs and sSTB-EVs respectively (Supplemental table 4 [↗](#)).

Discussion

Principal findings

In our analysis, PE and NP have different placenta and STB-EV proteomes. Four STB-EV biomarkers—filamin B, collagen 17A1, pappalysin-A2, and scavenger Receptor Class B Type 1—were verified for differential abundance. In silico investigation revealed molecular pathways such as abnormal protein metabolism that may contribute to PE's clinical and pathological symptoms and inform future research.

Results in the context of what is known

Our analysis found chloride intracellular channel 3 (CLIC3) to be the only protein differentially abundant among all sample types. CLIC3 is expressed in the placenta throughout pregnancy (Money et al., 2007 [↗](#)) and is abundant in PE placentas compared to NP, a finding corroborated by our results. However, it has not previously been described in terms of STB-EVs. CLIC3 also recycles activated integrins back to the plasma membrane and facilitates cell migration and invasion (Dozynkiewicz et al., 2012 [↗](#)) a process that has been identified as abnormal in PE (KHONG et al., 1986).

We found *COL17A1* to be more abundant in PE m/lSTB-EVs but not detectable in the placenta in both normal and PE. Although collagen 17 has not been described in preeclampsia, a protein of the same family, collagen 1 is deposited in higher amounts in the PE placenta and can induce preeclampsia-like symptoms by suppressing the proliferation and invasion of trophoblasts. This suppression was reversible by treating with ERK and B-catenin agonists (Feng et al., 2021 [↗](#)). Likewise, *PAPP-A2* was significantly more abundant in the PE placenta and m/lSTB-EVs. *PAPP-A2* cleaves insulin-like growth factor binding protein (IGFBP-5 and, to a lesser extent, IGFBP-3). *PAPP-A2*'s mRNA and protein are differentially expressed in the placenta and maternal serum in PE patients (Whitehead et al., 2013 [↗](#)).

Filamin B was significantly more abundant in PE m/ISTB-EVs and sSTB-EVs compared to NP. Filamin B participates in cellular structural mechanics and signal transduction by interacting with ion channels, signaling molecules, transmembrane proteins, and transcription factors (Zhou et al., 2010). It also suppresses tumor growth and metastasis (Iguchi et al., 2015). Interestingly, in contrast to our study Wei et al described Filamin B as being up regulated in the placenta (Wei et al., 2019). Wei et al used glyceraldehyde 3-phosphate dehydrogenase (GAPDH) as an internal reference while we used total protein normalization to Amido black to quantify and compare protein expression. GAPDH is an unreliable internal reference protein, particularly in preeclampsia (Lanoix et al., 2012) and this may explain these discordant findings. In addition, in our proteomics data, we found GAPDH to be among the most differentially expressed proteins.

Likewise, we identified scavenger receptor class B, type 1 (SCARB1/SR-BI) to be significantly increased in PE placenta, m/l and sSTB-EVs. SCARB1/SR-BI, is most abundant in the adrenal glands, placenta, liver, and brain (Ganesan et al., 2016; Shen et al., 2016). SR-BI facilitates the uptake of cholesterol esters from high-density lipoproteins and lipid-soluble vitamin and transthyretin-bound thyroid hormone by placental trophoblast cells (Landers et al., 2018).

In terms of potential mechanisms of preeclampsia, we found no overlap in the biological processes and KEGG pathways among the three sample sub-type. In the placenta, gene ontology biological processes (GO: BP) involved in *neurotransmitter secretion and transport* were overrepresented while *platelet activation*, *MAPK* and *Rap 1 signaling* pathways were among the detected KEGG pathways. In m/ISTB-EVs, *protein modification process*, and *decreased oxygen level responses* were among the perturbed GO: BP terms. *Alzheimer's* and *prion disease* were among the identified KEGG pathways. In sSTB-EVs, *post-translational modification processes* and *endoplasmic reticulum protein processing* were the principal GO: BP terms and KEGG pathways.

Recent research has shown that ischemic hypoxia and the release of proinflammatory cytokines in PE can cause protein misfolding and initiate endoplasmic reticulum (ER) stress due to hypoxia-reoxygenation damage to the endoplasmic reticulum (Gathiram & Moodley, 2016). PE can also cause posttranslational modifications to proteins, such as changing the isoelectric point, which results in different S-nitrosylation outcomes in placental proteins (Zhang et al., 2011). It is thought that the accumulation of these aggregates of unfolded protein response (UPR) or misfolded proteins contributes to the pathophysiology of PE (Gathiram & Moodley, 2016). Other processes and KEGG pathways have been previously described in preeclampsia, while others found in our study are new and may warrant further research (Lee et al., 2020; Wan Shumei, Peng Ping, Qiao Lin, 2019).

Ideas and Speculation

STB-EVs are liquid biomarkers with real-time information from the damaged placenta in PE due to their placenta-specificity. It would be interesting to test maternal plasma or serum samples for the four STB-EV indicators reported in our study.

It is unclear which comes first: misfolded proteins depositing in trophoblasts and preventing normal invasion, causing ischemia and endoplasmic reticulum stress, which leads to defects in trophoblast invasion, oxidative stress, and endothelial cell dysfunction, or the trophoblast invasion defects, oxidative stress, and endothelial cell dysfunction, or the misfolded proteins because of faulty invasion and oxidative stress. Further studies exploring these PE pathogenic processes would be intriguing.

Strengths and limitations

Our study explored the difference in the proteome between PE and NP by analyzing the placenta and its extracellular vesicles. This study is one of the few to do so using extracellular vesicles obtained by a physiologic technique, the *ex-vivo* dual lobe placenta perfusion. However, the

control population used for this study was not gestationally age-matched because it is impossible to obtain the ideal control for early onset PE patients. Also, our sample size was small (n of 12).

Conclusion

Our study may have found novel STB-EV-bound protein indicators that are significantly more abundant in PE than normal. Since STB-EVs are present in the circulation from early pregnancy to term and are released more in PE, these STB-EV carried proteins may help with earlier diagnosis and mechanistic insights.

Data Availability

All data produced are available online at

<https://data.mendeley.com/datasets/j2x4h9ddcj/draft?a=8ccd383c-c6e5-495d-b66b-29c691608995> 

Acknowledgements

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Competing interests

The authors declare no competing interests.

Supplemental Material

Supplemental Methods

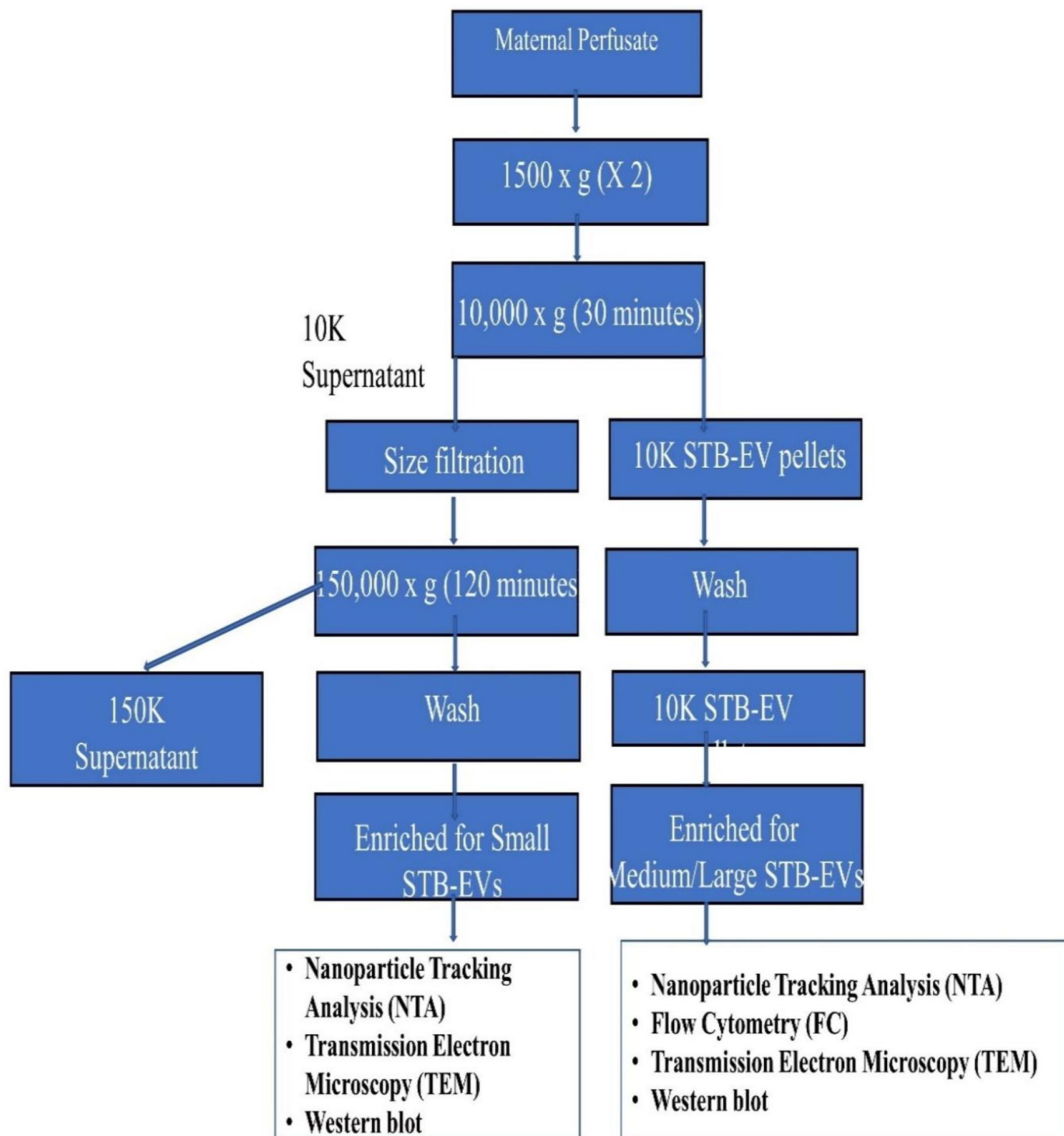
Supplemental table S1-4

Supplemental figure S1

List of differentially abundant proteins and functional enrichment analysis: <https://data.mendeley.com/datasets/j2x4h9ddcj/draft?a=8ccd383c-c6e5-495d-b66b-29c691608995> 

Supplemental data

Supplemental methods



Supplemental figure 1.

Flow chart illustrating the steps involved in characterizing m/ISTB-EVs and sSTB-EVs obtained via differential ultracentrifugation (10,000 and 150,000 g) of maternal perfusate. Nanoparticle trafficking analysis, transmission electron microscopy, flow cytometry, and western blot were used to characterize the pellets from both spins.

Transmission electron microscopy

The Sir William Dunn School of Pathology was contracted to perform transmission electron microscopy. STB-EV pellets were diluted with fPBS to produce STB-EV solutions with concentrations ranging from 0.1 to 0.3 g/l. For 2 minutes, ten microliter of the STB-EV pellet solution was applied to freshly glowing discharged carbon formvar 300 mesh copper grids, blotted with filter paper, stained with 2% uranyl acetate for 10 seconds, blotted, and air-dried. The grid's STB-EV pellets are negatively stained to increase the contrast between the STB-EV pellets and the background. The grids were imaged with a Gatan OneView CMOS camera on an FEI Tecnai 12 TEM at 120 kV.

Nanoparticle tracking analysis

The Nanosight NS500 (instrument equipped with a 405 nm laser [Malvern UK]), sCMOS camera, and nanoparticle tracking analysis (NTA) software version 2.3, Build 0033 (Malvern UK) system was used for the analysis. Instrument performance was tested with silica 100 nm microspheres prior to sample analysis (Polysciences, Inc.). The samples were diluted in fPBS to a concentration to 1/100,000 based on the starting concentration. The samples were automatically injected into the sample chamber with a 1 ml syringe using the EV measurement script: prime, delay of 5, capture of 60, and repeat of 4. Camera images of the analyzed samples were captured at a level of 12. (Camera shutter speed; 15 ms and Camera gain; 350). NTA post-acquisition settings were optimized and maintained constant across samples. Each video recording was analyzed to determine the size and concentration profile of STB-EV.

Western blotting

Bioinformatic analysis of proteomic data from placenta tissue, medium/large and small STB-EVs

Persus (Max Planck Institute of Biochemistry) was used for the analysis, along with the accompanying documentation and tutorials. To remove invalid data, we pre-processed the raw data by log transforming and filtering. Missing values were imputed at random from a normal distribution using the following parameters: width = 0.3 and downshift = 1.8. To ensure conformity to the normal distribution, the underlying distribution was visually inspected with a histogram before and after missing data imputation. The data was then transformed further by deducting each (transformed) value from the highest occurring protein expression value.

Principal component analysis (PCA), heatmaps, and Pearson correlation matrices were used to further investigate the data. The data was analyzed using the correlation index and hierarchical clustering. A two-sample independent student t-test was used to assess differential expression. Multiple testing was corrected using permutation-based false discovery rate (FDR) with the following parameters, with significance set at less than 0.05.

Functional enrichment of differentially expressed proteins (DEPs) in preeclampsia (PE)

Buffers	Power of hydrogen (pH)	Constituents
4X Laemmli reducing buffer	6.8	9 parts 4 X Laemmli buffer 1 part 2-beta-mercaptoethanol
4X Laemmli non-reducing buffer		4 X Laemmli buffer
Anode buffer 1	10.4	300 mM Tris base (36.34 g/L) 100 % methanol (20 mL/L) 980 ml of ddH ₂ O
Anode buffer 2	10.4	25 mM Tris base (3.0 g/L) 100 % methanol (20 mL/L) 980 ml of ddH ₂ O
Cathode buffer	9.4	25 mM Tris base (3.0 g/L) 40 nM 6-aminocaproic acid (5.2 g/L) 100 % methanol (20 mL/L) 980 mL of dH ₂ O
10 X TBS	8	NaCl (87.6g) Tris base (12.1g) 700 ml of ddH ₂ O
TBST	8	10 X TBS (100 mL) 0.1 % Tween-20 (1 mL) 900 mL of ddH ₂ O
5% Milk TBST		5g of Blotto in 100 mL of TBST

Supplemental table 1.

Reagents used for western blot.

Antibodies	Concentration	Dilution	Antigen	Specificity	Manufacturer
Anti-PLAP (NDOG 2)	1.6 µg/µl	1/1000	PLAP	STB-EV	In house antibody
Anti-CD63	200 µg/µl	1/1000	CD63	STB-EV	Santa Cruz Biotechnology
Anti-ALIX	200 µg/µl	1/1000	ALIX	S STB-EV	Cell Signalling
Anti-Cytochrome C	200 µg/µl	1/500	Cytochrome C	Placenta homogenate	Santa Cruz Biotechnology
Anti-SR-BI	0.35 µg/µl	1/1000	SR-BI	N/A	Abcam
Anti-Filamin B	1.64 µg/µl	1/1000	Filamin B	N/A	Insight Biotechnology
Anti-PAPP-A2	1 µg/µl	1/2000	PAPP-A2	N/A	Abcam
Anti-Collagen 17 A1	1 µg/µl	1/1000	Collagen 17 A1	N/A	Abcam
Polyclonal goat-anti-mouse/rabbit immunoglobulin HRP		1/2000	Mouse and Rabbit Immunoglobulins	N/A	Dako UK Ltd

Supplemental table 2.

Antibodies used for western blot.

Biological Process (BP)	Adjusted P Values	No of Proteins in BP terms	No of DEP queried	No of DEP found in BP terms
Negative regulation of neurotransmitter secretion	0.02	2	14	2
Negative regulation of synaptic vesicle exocytosis	0.02	2	14	2
Negative regulation of neurotransmitter transport	0.05	3	14	2
Response to decreased oxygen levels	0.00	84	293	30
Cellular response to decreased oxygen levels	0.00	59	293	24
Cellular ketone metabolic process	0.00	65	293	26
<i>Peptidyl-asparagine modification</i>	<i>0.02</i>	<i>10</i>	<i>73</i>	<i>5</i>
<i>Protein N-linked glycosylation via asparagine</i>	<i>0.02</i>	<i>10</i>	<i>73</i>	<i>5</i>
<i>Protein N-linked glycosylation</i>	<i>0.02</i>	<i>16</i>	<i>73</i>	<i>6</i>

Supplemental table 3

The top three functionally enriched gene ontologies are: biological process (GO: BP) Placenta (bold font), medium/large STB-EVs (normal font), and small STB-EVs (italics).

KEGG Pathways (KP)	Adjusted P Values	No of Proteins in KP terms	No of DEP queried	No of DEP found in KP terms
Neurotrophin signalling pathway	0.00	13	14	3
Long-term potentiation	0.01	6	14	2
Renal cell carcinoma	0.02	11	14	2
Proteasome	0.00	35	293	17
Spinocerebellar ataxia	0.00	48	293	21
Alzheimer disease	0.01	94	293	29
<i>Protein processing in endoplasmic reticulum</i>	<i>0.02</i>	<i>60</i>	<i>73</i>	<i>10</i>

Supplemental table 4

The top three functionally enriched KEGG Pathways in the placenta, medium/large STB-EVs, and small STB-EVs (only 1 KEGG). Placenta (bold font), medium/large STB-EVs (normal font), and small STB-EVs (italics)

Major Resources Table

Target antigen	Vendor or Source	Catalog #	Working concentration	Dilution
PLAP	In house antibody		1.6 µg/µl	1/1000
CD63	Santa Cruz Biotechnology	sc-365604	200 µg/µl	1/1000
ALIX	Cell Signalling	#2171	200 µg/µl	1/1000
Cytochrome C	Santa Cruz Biotechnology	sc-13560	200 µg/µl	1/500
SR-BI	Santa Cruz Biotechnology	ab52629	0.35 µg/µl	1/1000
Filamin B	Insight Biotechnology	GTX387	1.64 µg/µl	1/1000
PAPP-A2	Abcam	ab59100	1 µg/µl	1/1000
Collagen 17A1	Abcam	Ab28440	1 µg/µl	1/2000
Mouse Immunoglobulins	Dako UK Ltd	P044701	1 µg/µl	1/2000
Rabbit Immunoglobulins	Dako UK Ltd	P044801	1 µg/µl	1/2000

Markers	Fluorochromes	Clone	CAT	Isotype	Dilution / Concentration
CD41	PE Vio770	REA386	130-105-562	REA	1 in 50 for dump channel analysis.
CD235	PE Vio770	RAE-175	130-100-258	REA	
HLA Class-I (ABC)	PE Vio770	REA230	130-101-460	REA	
HLA Class-II (DRDPDQ)	PE Vio770	RAE-332	130-104-828	REA	
REA control				REA	
CD41	Pacific blue	HIP8	303713	Mouse IgG1	1 in 100
CD235	Pacific blue	HI264	349107	Mouse IgG2a	1 in 100
HLA Class-I (ABC)	Pacific blue	W6/32	311417	Mouse IgG2a	1 in 100
PLAP	PE	Mouse mAb	N/A	IgG1	0.2mg/ml

IgG2a isotype control	Pacific blue	MOPC-173	981904	IgG2 a	1 in 500
IgG1 isotype control	Pacific blue	MOPC-21	400131	IgG1	1 in 200
bioM	FITC	NA	NA	NA	0.5-1nM

(continued)

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Reviewer #1 (Public Review):

The authors primary objective in this study was to identify differences between patients with preeclampsia and normal patients with respect to the placental syncytiotrophoblast extracellular vesicle proteome.

One of the strengths of this study is that it is one of only a few studies that investigated the difference in proteome between patients with preeclampsia and those with normal pregnancies using placental extracellular vesicles obtained by an ex-vivo dual lobe placenta perfusion technique.

The main weaknesses of this study are:

1. The small sample size in that there were only 12 cases.
2. The study patients and control population of normal pregnancies were not matched for gestational age at delivery.

The authors were able to achieve their study aims and the results support their conclusions.

These findings could be used in future studies of the disease mechanisms and as biomarkers for prediction of preeclampsia. As such, they may be very useful for the identification of women at risk for preeclampsia well before the onset of disease.

Reviewer #2 (Public Review):

Summary:

Preeclampsia is a disorder of pregnancy that affects 4-5% of pregnancies worldwide. Identifying this condition early is clinically relevant as it will help clinicians to make management decisions to prevent adverse outcomes. The placenta holds a key to many pregnancy-related pathologies including preeclampsia and studies have shown many differences in the placenta of women with preeclampsia as compared to controls. However as the placenta cannot be collected directly during pregnancy, the exosomes secreted by it are considered a good alternative to tissue biopsy. In this study, the authors have compared the proteins in different sizes of exosomes from the placenta of women with and without preeclampsia. The idea is to eventually use these as biomarkers for early detection of preeclampsia.

Strengths:

The novelty factor of this study is the use of two different-sized exosomes which has not been achieved earlier.

Weaknesses:

There is already enough information about the differences in exosome contents from the placentas of women with and without preeclampsia. There are some issues with the methods which may influence the outcomes of the data.

The patient population described in the methods section is of HELLP syndrome while the title and the manuscript describe preeclampsia. While it is an important life-threatening condition to address, it is extremely rare and needs careful assessment by clinicians in terms of patient characteristics and outcomes measured.

The study measured the proteins at only a single time point after the disease has already occurred. However, the placenta is an ever-changing tissue throughout pregnancy and different proteins can come up at different times in pregnancy. Thus serial measurements are necessary and a single time point measurement like that done here does little value addition. Unfortunately, this site has not validated the identified biomarkers in plasma or circulating placental exosomes from women with and without preeclampsia. Thus the validity of these findings in real-life situations can not be judged.